

both for simulation and real-time applications. In this article, the simulation mode is explored for the animation of autonomous cars.

Installation of Node-RED and associated components

The latest version of Node.js can be installed from the URL <https://nodejs.org/en/download/>. The checkout wizard for the same can be completed with full administration permissions sanctioned.

To check if Node.js has been installed successfully, without any issues, the following command is used. This also checks the version installed.

```
node -v
```

After its setup, the node package manager (NPM) is installed by typing the command given below:

```
npm install -g node-red
```

To check the installation status or version of Node-RED, give the following command. This command shows the verbose output while the server is being started, as shown in Figure 1.

```
node-red -v
```

After the installation, to view the default Node-RED screen, the website must be opened with the URL <http://127.0.0.1:1880> using the browser. This URL address notifies that we have the Node-RED server running on the local machine on port 1880.

To install the SVG graphics node, the following command is to be given:

```
npm install node-red-contrib-ui-svg
```

Architecture for animation in autonomous cars

A new flow is created first, and an SVG graphics node is added. Later, a car is drawn in the SVG editor using elements like lines and

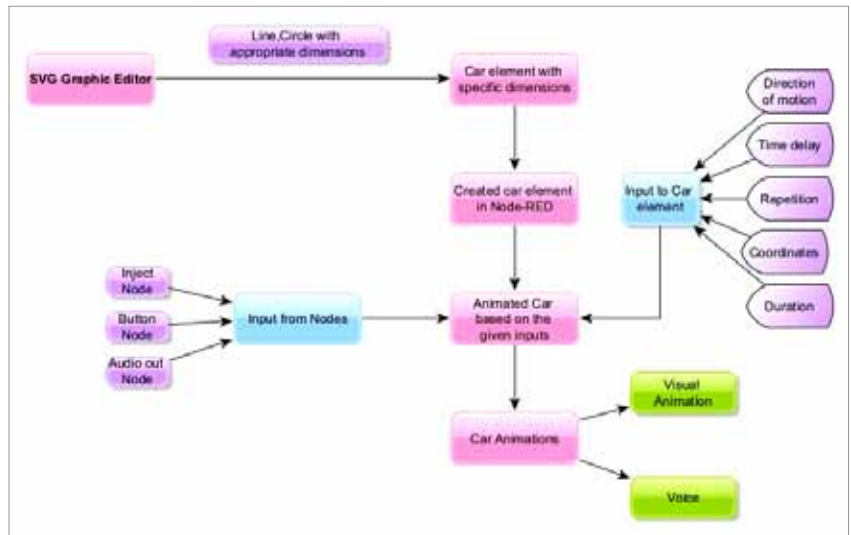


Figure 1: Command to start the Node-RED server

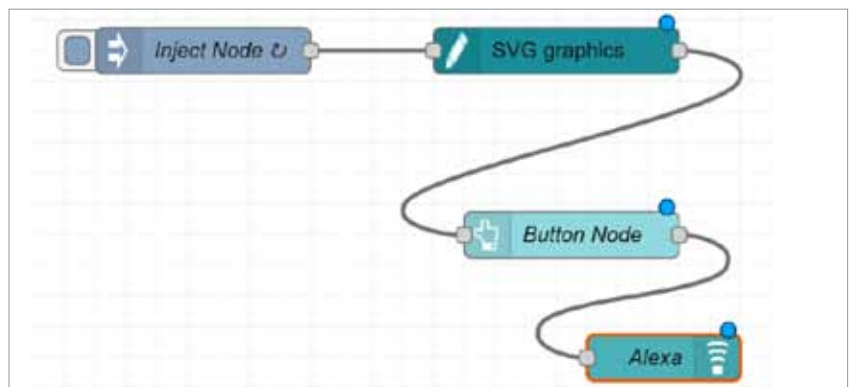


Figure 2: Architecture diagram for autonomous car animation

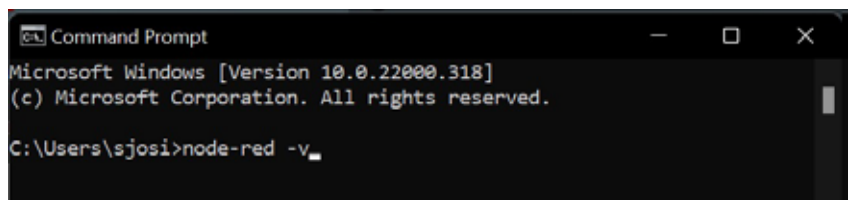


Figure 3: Node-RED flow for autonomous car design

circles. Next, to add animation, there should be specific parameters like direction of motion, time delay between movements, number of times the movement has to be repeated, coordinates of movement and the duration of movement. After creating the animated car, configure the 'button', 'audio out' and the 'inject' nodes. A connection is made between the SVG graphics, inject button and

audio out nodes in the Node-RED flow. A stationary car will be visible in the Node-RED dashboard (output window) and animation is triggered once the button is pressed. The entire workflow is depicted in Figure 2.

Design for autonomous car animation

The steps for installing and opening the Node-RED interface are